An essential difference between the granite analogy and our real concern with the spatial fabric is that there are ways in which the microscopic discreteness of the granite's surface can be exposed: Finer, more precise probes than our fingers can be used. An electron microscope has the ability to resolve surface features to less than a millionth of a centimeter; this is sufficiently small to reveal the numerous surface imperfections. By contrast, in string theory there is no way to expose the sub-Planck-scale "imperfections" in the fabric of space. In a universe governed by the laws of string theory, the conventional notion that we can always dissect nature on ever smaller distances, without limit, is not true. There is a limit, and it comes into play before we encounter the devastating quantum foam of Figure 5.1. Therefore, in a sense that will be made more precise in later chapters, one can even say that the supposed tempestuous sub-Planckian quantum undulations do not exist. A positivist would say that something exists only if it can—at least in principle—be probed and measured. Since the string is supposed to be the most elementary object in the universe and since it is too large to be affected by the violent sub-Planck-length undulations of the spatial fabric, these fluctuations cannot be measured and hence, according to string theory, do not actually arise.

A Sleight of Hand?

This discussion may leave you feeling dissatisfied. Instead of showing that string theory tames the sub-Planck-length quantum undulations of space, we seem to have used the string's nonzero size to skirt the whole issue completely. Have we actually solved anything? We have. The following two points will serve to emphasize this.

First, what the preceding argument implies is that the supposedly problematic sub-Planck-length spatial fluctuations are an artifact of formulating general relativity and quantum mechanics in a point-particle framework. In a sense, therefore, the central conflict of contemporary theoretical physics has been a problem of our own making. Because we previously envisioned all matter particles and all force particles to be pointlike objects with literally no spatial extent, we were obligated to consider properties of the universe on arbitrarily short distance scales. And on the tiniest of distances we ran into seemingly insurmountable problems. String theory tells us that we